

DevOps Project Report:

Scientific Calculator

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1. Introduction to DevOps

DevOps is a set of practices that combines software development (Dev) and IT operations (Ops) to improve collaboration and automate the software delivery process. The main goal of devOps is to develop, test, and release software faster and more reliably. It uses practices like Continuous Integration (CI), Continuous Deployment (CD) and automation to reduce errors and improve efficiency in the software development lifecycle.

Traditional software development relies on manual build, testing, and development processes, which can be slow, inconsistent, and prone to human errors. DevOps addresses these issues by automating repetitive tasks, enabling faster feedback, improving code quality, and ensuring consistent application deployment.

This project mainly focuses on building **Scientific Calculator** using modern DevOps practices. The application is built using Java and supports the following operations:

- **Square root function** - $\sqrt{x}$
- **Factorial function** - $x!$
- **Natural logarithm (base e)** - $\ln(x)$
- **Power function** - x^b

2. Tools used

Tools	Purpose
Git & GitHub	Source Code Management
Java	Application logic
JUnit	Unit Testing
Maven	Build Automation & dependency management
Jenkins	Continuous Integration & Delivery (CI/CD)
Docker	Containerization
Docker Hub	Image registry
Ansible	Automated deployment

3. Project Setup

Git Commands

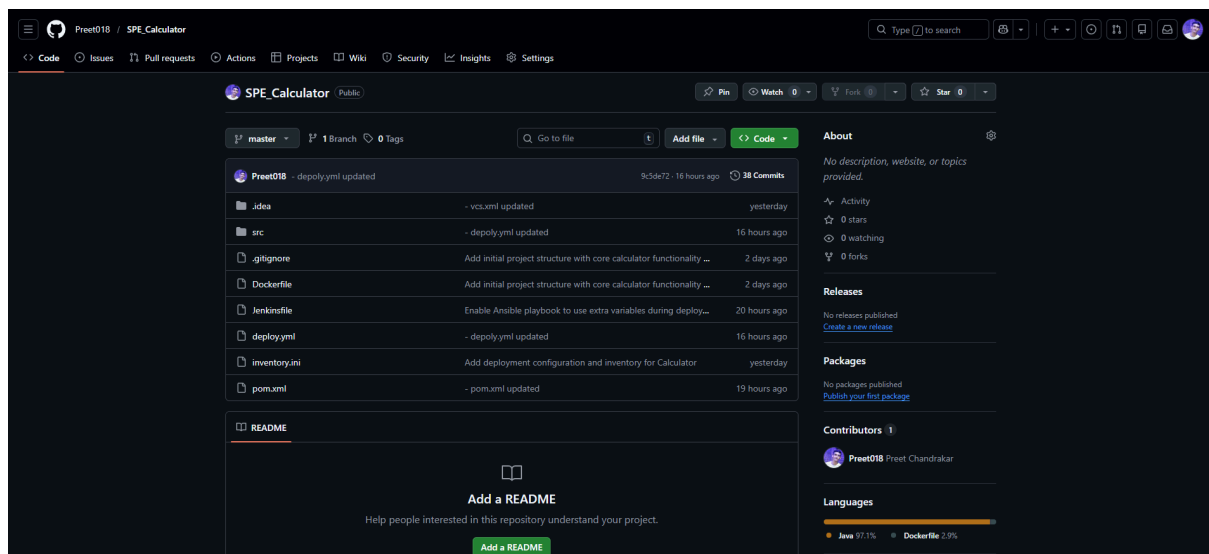
git clone https://github.com/Preet018/SPE_Calculator.git

git add .

git commit -m "Changes"

git push origin master

Screenshot 1: GitHub Repository



4. Testing with JUnit

Test cases:

- Valid edge cases (e.g. factorial of negative number, square root of negative number, etc)
- Parameterized tests for multiple inputs.

Code Snippet: CalculatorTest.java

```
1  import org.example.Calculator;
2
3  import org.junit.jupiter.api.Test;
4  import static org.junit.jupiter.api.Assertions.*;
5
6  public class CalculatorTest {
7      // Square Root Function
8      @Test
9      public void testSquareRoot() {
10         assertEquals( expected: 4.0, Calculator.squareRoot( x: 16), delta: 0.0001);
11         assertEquals( expected: 5.0, Calculator.squareRoot( x: 25), delta: 0.0001);
12     }
13
14     @Test
15     public void testSquareRootNegative() {
16         assertThrows(IllegalArgumentException.class, () -> {
17             Calculator.squareRoot( x: -9);
18         });
19
20         assertThrows(IllegalArgumentException.class, () -> {
21             Calculator.squareRoot( x: -15);
22         });
23     }
24
25     // Factorial Function
26     @Test
27     public void testFactorial() {
28         assertEquals( expected: 120.0, Calculator.factorial( x: 5));
29         assertEquals( expected: 1.0, Calculator.factorial( x: 0));
30     }
31
```

```
32
33     @Test
34     public void testFactorialNegative() {
35         assertThrows(IllegalArgumentException.class, () -> {
36             Calculator.factorial( x: -3);
37         });
38
39         assertThrows(IllegalArgumentException.class, () -> {
40             Calculator.factorial( x: -10);
41         });
42     }
43
```

```

43     @Test
44     public void testFactorialTooLarge() {
45         assertThrows(IllegalArgumentException.class, () -> {
46             Calculator.factorial(x: 45);
47         });
48     }
49
50     // Natural Log Function
51     @Test
52     public void testNaturalLog() {
53         assertEquals(Math.log(10), Calculator.naturalLog(x: 10), delta: 0.0001);
54         assertEquals(Math.log(1), Calculator.naturalLog(x: 1), delta: 0.0001);
55     }
56
57     @Test
58     public void testNaturalLogInvalid() {
59         assertThrows(IllegalArgumentException.class, () -> {
60             Calculator.naturalLog(x: 0);
61         });
62
63         assertThrows(IllegalArgumentException.class, () -> {
64             Calculator.naturalLog(x: -18);
65         });
66     }
67

```

```

68     // Power Function
69     @Test
70     public void testPower() {
71         assertEquals(expected: 32.0, Calculator.power(x: 2, b: 5), delta: 0.0001);
72         assertEquals(expected: 27.0, Calculator.power(x: 3, b: 3), delta: 0.0001);
73         assertEquals(expected: 1.0, Calculator.power(x: 5, b: 0), delta: 0.0001);
74     }
75
76     @Test
77     public void testPowerInvalid() {
78         assertThrows(IllegalArgumentException.class, () -> {
79             Calculator.power(x: 0, b: -2);
80         });
81
82         assertThrows(IllegalArgumentException.class, () -> {
83             Calculator.power(x: 0, b: -5);
84         });
85     }
86 }
87

```

Screenshot 2: JUnit Test Results

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running CalculatorTest
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.062 s -- in CalculatorTest
[INFO] Results:
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ Calculator ---
[INFO] Building jar: C:\Users\chand\Downloads\IIIT Bangalore Stuffs\Term 2\SPE\Mini Project\Calculator\target\Calculator-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.868 s
[INFO] Finished at: 2026-03-06T15:13:49+05:30
[INFO] -----

```

5. Build with Maven

pom.xml configuration:

- Added dependencies for JUnit and Maven plugin.
- Configured <mainClass> for executable JAR.

Maven build command:

mvn clean package

Screenshot 3: Maven Build Output

```
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.example:Calculator >-----
[INFO] Building Calculator 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ Calculator ---
[INFO] Deleting C:\Users\chand\Downloads\IIIT Bangalore Stuffs\Term 2\SPE\Mini Project\Calculator\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ Calculator ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ Calculator ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to C:\Users\chand\Downloads\IIIT Bangalore Stuffs\Term 2\SPE\Mini Project\Calculator\target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ Calculator ---
[INFO] skip non existing resourceDirectory C:\Users\chand\Downloads\IIIT Bangalore Stuffs\Term 2\SPE\Mini Project\Calculator\src\test\resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ Calculator ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to C:\Users\chand\Downloads\IIIT Bangalore Stuffs\Term 2\SPE\Mini Project\Calculator\target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ Calculator ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running CalculatorTest
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.062 s -- in CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ Calculator ---
[INFO] Building jar: C:\Users\chand\Downloads\IIIT Bangalore Stuffs\Term 2\SPE\Mini Project\Calculator\target\Calculator-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.868 s
[INFO] Finished at: 2026-03-06T15:13:49+05:30
[INFO] -----
```

6. Continuous Integration with Jenkins

Jenkins Pipeline:

- **Stages:** Checkout → Build → Test → Build Docker Image → Push Docker Image → Ansible Deployment

Code Snippet: Jenkinsfile

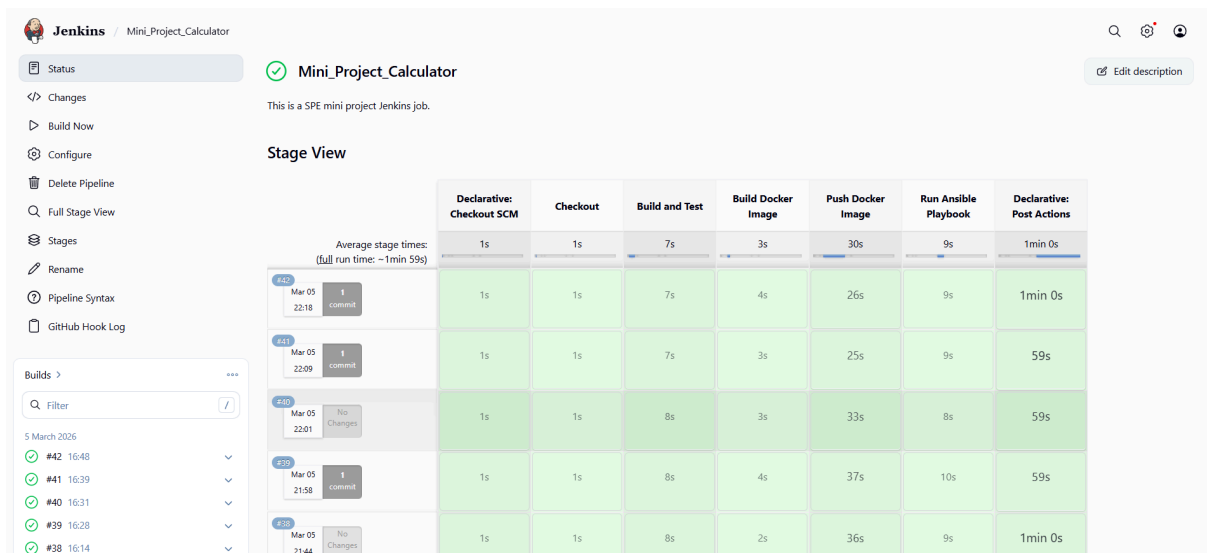
```
1 pipeline {
2     agent any
3     environment {
4         DOCKER_IMAGE_NAME = 'scientific-calculator'
5         GITHUB_REPO_URL = 'https://github.com/Preet018/SPE_Calculator.git'
6         DOCKER_USERNAME = 'preet018'
7         IMAGE_TAG = '0.0.1'
8     }
9     stages {
10        stage('Checkout') {
11            steps {
12                script {
13                    git branch: 'master', url: "${GITHUB_REPO_URL}"
14                }
15            }
16        }
17        stage('Build and Test') {
18            steps {
19                script {
20                    sh 'mvn clean install'
21                }
22            }
23        }
24        stage('Build Docker Image') {
25            steps {
26                script {
27                    docker.build("${DOCKER_IMAGE_NAME}:${IMAGE_TAG}", '.')
28                    // Optionally, also tag the image as "latest" for consistency
29                    docker.build("${DOCKER_IMAGE_NAME}:latest", '.')
30                }
31            }
32        }
33        stage('Push Docker Image') {
34            steps {
35                script {
36                    docker.withRegistry('', 'DockerHubCred') {
37                        // Tag the image with the appropriate version and latest
38                        sh "docker tag ${DOCKER_IMAGE_NAME}:${IMAGE_TAG} ${DOCKER_USERNAME}/scientific-calculator:${IMAGE_TAG}"
39                        sh "docker tag ${DOCKER_IMAGE_NAME}:${IMAGE_TAG} ${DOCKER_USERNAME}/scientific-calculator:latest"
40
41                        // Push both the versioned tag and latest tag
42                        sh "docker push ${DOCKER_USERNAME}/scientific-calculator:${IMAGE_TAG}"
43                        sh "docker push ${DOCKER_USERNAME}/scientific-calculator:latest"
44                    }
45                }
46            }
47        }
48        stage('Run Ansible Playbook') {
49            steps {
50                script {
51                    ansiblePlaybook(
52                        playbook: 'deploy.yml',
53                        inventory: 'inventory.ini',
54                        extras: '-K'
55                    )
56                }
57            }
58        }
59    }
60 }
```

```

60     post {
61         success {
62             echo 'Pipeline successfully completed!'
63             emailx(
64                 to: 'chandrakarpreet.1100@gmail.com',
65                 subject: 'Build Success: Scientific Calculator',
66                 body: 'The Jenkins pipeline for the Scientific Calculator project has completed successfully.'
67             )
68         }
69         failure {
70             echo 'Pipeline failed!'
71             emailx(
72                 to: 'chandrakarpreet.1100@gmail.com',
73                 subject: 'Build Failure: Scientific Calculator',
74                 body: 'The Jenkins pipeline for the Scientific Calculator project has failed.'
75             )
76         }
77     }
78 }
79

```

Screenshot 4: Full jenkins pipeline view



7. Containerization with Docker

Code Snippet: Dockerfile

```

1 FROM amazoncorretto:17
2
3 WORKDIR /app
4
5 # Copy the built JAR file from the previous stage
6 COPY target/Calculator-1.0-SNAPSHOT.jar /app/calculator.jar
7
8 # Start up cmd to run the application
9 ENTRYPOINT ["java", "-jar", "calculator.jar"]

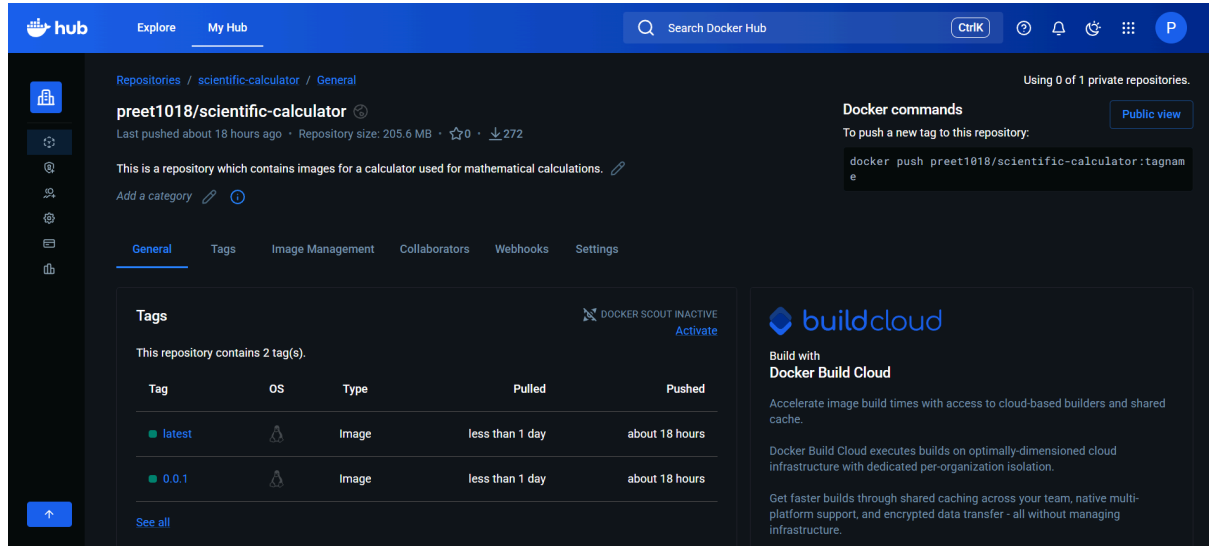
```

Docker commands:

```
docker build preet1018/scientific-calculator:0.0.1 .
```

```
docker push preet1018/scientific-calculator:0.0.1
```

Screenshot 5: Docker image(s) on Docker Hub



8. Deployment with Ansible

Code Snippet: `deploy.yml`

```
1 ---
2 - name: Deploy Calculator
3   hosts: localhost
4   become: yes # Enable sudo
5   become_user: root # Run tasks as root
6   tasks:
7     - name: Install Docker
8       apt:
9         name: docker.io
10        state: present
11
12    - name: Start Docker service
13      service:
14        name: docker
15        state: started
16
17    - name: Pull Docker image
18      docker_image:
19        name: preet1018/scientific-calculator:0.0.1
20        source: pull
21
22    - name: Run container
23      docker_container:
24        name: calculator
25        image: preet1018/scientific-calculator:0.0.1
26        state: started
27        restart_policy: always
28        interactive: true
```


Code Snippet: inventory.ini

```
1 [local]
2 localhost ansible_connection=local ansible_python_interpreter=/usr/bin/python3 ansible_become_pass=9589
```

Screenshot 6: Ansible Deployment Output

```
chand@Preet:/mnt/c/Users/chand/Downloads/IIIT Bangalore Stuffs/Term 2/SPE/Mini Project/Calculator$ ansible-playbook -i inventory.ini deploy.yml
PLAY [Deploy Calculator] *****
TASK [Gathering Facts] *****
ok: [localhost]
TASK [Install Docker] *****
ok: [localhost]
TASK [Start Docker service] *****
ok: [localhost]
TASK [Pull Docker image] *****
ok: [localhost]
TASK [Run container] *****
ok: [localhost]
PLAY RECAP *****
localhost : ok=5  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

9. Application Screenshots

Screenshot 7: Application's Command Line Interface (CLI)

```
chand@Preet:/mnt/c/Users/chand/Downloads/IIIT Bangalore Stuffs/Term 2/SPE/Mini Project/Calculator$ docker run -it preet1018/scientific-calculator
Select an operation:
1. Square Root (/x)
2. Factorial (x!)
3. Natural Logarithm (ln(x))
4. Power (x^b)
5. Exit
1
Enter a number to find its square root:
4
Result: 2.0
Select an operation:
1. Square Root (/x)
2. Factorial (x!)
3. Natural Logarithm (ln(x))
4. Power (x^b)
5. Exit
2
Enter a number to find its factorial:
3
Result: 6
Select an operation:
1. Square Root (/x)
2. Factorial (x!)
3. Natural Logarithm (ln(x))
4. Power (x^b)
5. Exit
5
Exiting the calculator...
```

10. Conclusion

This project shows how to create and implement a **Scientific Calculator** application using DevOps practices. The project ensures that the application can be built, tested, packaged, and deployed consistently by incorporating automation into various phases of the software

development lifecycle. The development process is more efficient due to automated pipeline, which also reduces the possibility of human error and manual interactability.

Links:

- GitHub Repository:
https://github.com/Preet018/SPE_Calculator
- Docker Repository:
<https://hub.docker.com/repository/docker/preet1018/scientific-calculator/general>